

Is an automatic genre assignment feasible for the British Library 19th Century dataset?

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Abstract

This project aims to determine if there is a possibility to assign genre (and subgenre) using machine learning to a series of books from the collection of the British Library. In the dataset we have access to, there are 49 455 book titles, but only 19 309 are assigned a genre and subgenre (called type in the data frames). Our aim is to determine if there is a possibility, based on the known genres and types, to predict them for the unspecified books using machine learning. We'll explore several models based on support vector machines (SVM) to assess such a possibility.

Six different models, using different SVM techniques, were created to assign a genre and a subgenre. They succeeded in the task with different degrees of success but still made too many errors to be trusted for an automatic genre and subgenre assignment. However, having a look at how confident these models assign some genre/subgenre correctly could lead to a partial automation of the unspecified books assignment, decreasing the number of manual annotations.

1. Introduction

In the field of humanities, dealing with large amounts of data can be sometimes cumbersome. Some projects have come to light with the goal of benefitting from community expertise. The project initiated by Victoria Morris in 2005 with the 19th books collection from the British Library is one of these¹. The aim was to build a community-based metadata for books, and one of the aspects of this project was to assign a genre to the books. As it can sometimes be difficult to rely on a user-based approach to deal with data challenges, it proves to be interesting to investigate what machines can do.

¹ Van Strien, Daniel, Tolfo Giorgia, Morris, Victoria and Beelen, Kaspar. 2021. "Classifying 19th Century British Library books using Crowdsourcing and Machine Learning" in Living With Machines. © Copyright 2021 The Alan Turing Institute, British Library Board, Queen Mary University of London, University of Exeter, University of East Anglia and University of Cambridge. URL: <https://living-with-machines.github.io/genre-classification/intro.html>

The objective of this project is based on an issue already noticeable in the British Library 19th Century dataset filenames: some books lack a genre attribution. As it would be tedious to manually assign a genre to each book of this more than 30 000 rows document, it seems meaningful to investigate to what extent a computer would perform in this task. We don't claim to achieve such a goal, but we can at least explore some ways to assist this kind of practice.

2. Data preparation

The data we use comes from the 19th century archives of the British Library. It consists of 49 455 book entries, distributed over five Excel documents, four classified by genre (and two of them also containing subgenres, called type in the tables) and one attributed to books with an unspecified genre. The data was previously cleaned in a group work and can be found on this GitHub [repository](#).²

For the purpose of this project, two operations are still necessary. We first need to split the data into two sets, one containing [genre and subgenre](#), and one where they are [unspecified](#). We did this solely by dividing by initial documents names, so it may happen that books with no genre or type remain in the dataset, but they will be discarded later by the SVM models. Secondly, we chose to only work on English titles. This will on the one hand allow the models to perform better and on the other hand avoid language bias when assigning a genre.

We're doing these two steps using OpenRefine and will also discard columns we're not going to use. That is, columns with unique values or, on the contrary, with homogeneous cells. We could have filtered all of this in Python, but we find it easier to work on a dataset already set up for all our analyses, for readability purposes.

3. Methodology

As we said, we try to learn the classification of two kinds of values: genre and type. For the genre, we'll execute two types of models, both relying on support vector machines, which work well for classification based on small texts, like titles. The first one (*model 1*) will only take the titles into account, while the second one (*model 2*) will also incorporate other kinds of values such as the location of publication, the date, and the author.

For the type, besides the two kinds of architecture presented before (which become *model 3* and *model 4* for type attribution), we will also have to decide how to treat genre values. When it seems obvious that genre has a non-negligible significance while assessing a subgenre, this is only true for genres we know we can trust, that is, for the gold standard ones. If genres had to be previously predicted in order to be used for subgenre attribution, we can fear that previous errors will be reflected on the subgenre assignment. That's why we are running a model considering predicted genre (*model 5*) in contrast to a model using gold-labeled genre (*model 6*). We will then measure the efficiency of these two new models in comparison with *model 3* and *model 4*. The architecture of these models is briefly explained in section 4, where the reader can also have direct access to the code.

To compare these different models, we will focus on three types of analysis: a classification report, a confusion matrix, and a measure of uncertainty.

It is important to keep in mind that the conclusions we can draw from these numbers only make sense if the data is wholly correct and the data cleaning hasn't led to any mistake or inaccuracy. To be able to consider our analyses with a higher scientific level, we should have first performed a manual verification of a sufficient percentage of the data, and then drawn conclusions about the accuracy of the training dataset. This uncertainty about the data quality represents one more limitation to consider.

² URL : <https://github.com/petar-soldo/IDH2025-G5>

4. Training models

As said in the previous section, the analyses we want to conduct consider the implementation of six different models. Two of them, [model 1](#) and [model 3](#), rely on the same architecture and are the simplest as they just assess a value for genre or type solely based on the titles.

The others are multimodal models, as they also take into account the place of publication, the date and the author along with the titles. Two models of this sort are constructed: one for the genre ([model 2](#)), and one for the type ([model 4](#)).

Two more models will be created for the type assignment. The first ([model 5](#)) considers predicted values of genre when trying to assess values for type, while the second ([model 6](#)) takes the gold-labeled genres into account, and thus serves as standard.

To construct these models, generative AI (Google Gemini and OpenAI ChatGPT) was sometimes used, not for the global architecture of the models, but to help with specific code issues and redundancies.

All these models are built within Python and use support vector machines. The data is split into two sets, the first one for training and validation, the second one for testing. In each of these models, different analysis measures are computed such as a classification report, a confusion matrix and a measure of uncertainty.

5. Descriptive statistics³

Before viewing and interpreting the results, it can be useful to have a quick look at descriptive statistics of the data we will use.

We will work at most on a 18 087⁴ samples corpus, each representing a book with different kinds of metadata. As we can see on the plot below, *prose* represents more than half of the samples in our corpus. On the opposite, *music* only covers 4.9% of our dataset.

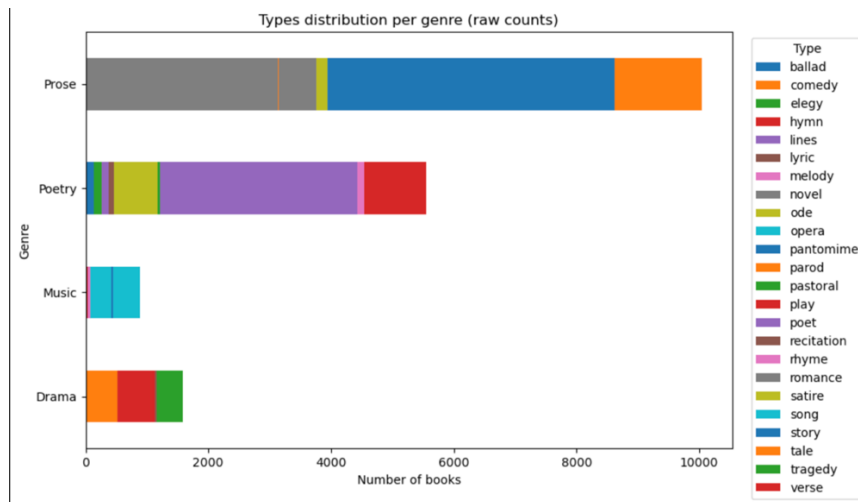


Figure 1: Distribution of genre and type in the corpus

Inside genre, some types are also mainly represented, this is the case for *story* and *novel* in *prose* genre, and for *poet* in *poetry* genre.

We can also observe an increasing number of publications per decade from the second half of the 18th century, culminating in the late 19th.

³ Other plots and numbers can be found in the notebook "[descriptive_statistics.ipynb](#)".

⁴ This number is surprisingly high, meaning that maybe too many books had previously been assigned with English language, testifying on a previous mistake being reflected on these analyses.

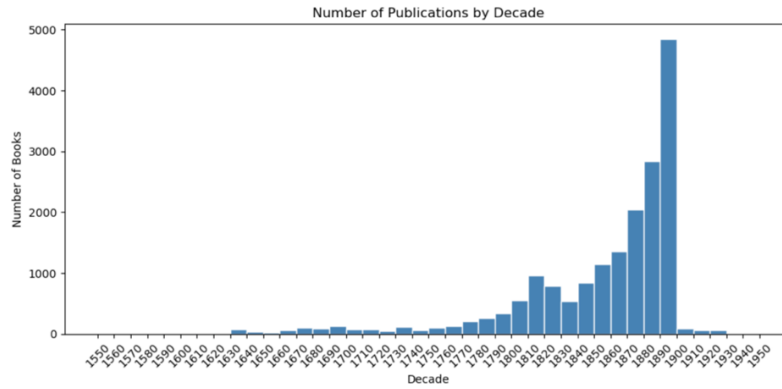


Figure 2: Number of books published per decade

We don't spend too much time analyzing these figures, as they're not the scope of our study. Still, it is important to keep them in mind, as they can explain some biases and overrepresentation in our machine learning models.

6. Results⁵

a. Genre: unimodal model (*model 1*) vs multimodal model (*model 2*)

To train classifying genre based on book titles, we have a total sample size of 18 087, from which 14 469 are used for training, and the remaining 3618 for testing. We are running a five-folds cross validation, with a final test accuracy of 93.2%.

Once we train genre on multiple variables, the number of samples drops to 15 091, with 12 072 left to training, and 3019 to testing. The final test accuracy is a bit higher, at 94.57%. Nevertheless, the improvement isn't that significant, and we need to keep in mind that it can be due to data leakage regarding authors already seen in the training phase or the number of books published in the 19th century.

In both cases, the confusion matrix shows good results, ranging from 87% to 97% in the first model, and from 81% to 99% in the second one. This latter achieves near perfection for *prose* assignment but performs with 6 points less for *music* one. It is a hint that such a multimodal model can suffer more from data sparsity, having better results for largely represented data. As we can see, *music* titles are more attributed to heavier genres in the second model.

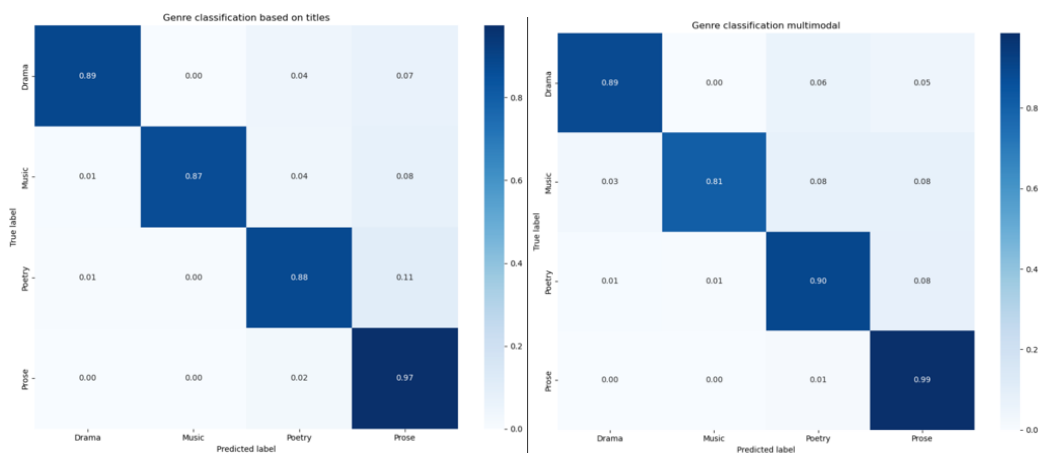


Figure 3: Confusion matrix for genre based on unimodal and multimodal models

⁵ All plots and figures aren't shown in this report but can be found in this GitHub [repository](https://github.com/Victor280296/IDH25-26_PersonalAssignment). URL: https://github.com/Victor280296/IDH25-26_PersonalAssignment

b. Type: unimodal model (*model 3*) vs multimodal (without genre) model (*model 4*)

To train for type classification based on book titles, we have again a total sample size of 18 087, from which 14 469 are used for training, and the remaining 3618 for testing. We are still running a five-folds cross validation, and the final test accuracy is 86.35%. When training on multiple variables, excluding genre at this point, the number of samples drops again to 15 091 and the final test accuracy is 88.04%, which is again a bit higher than for the unimodal model.

Once again, we can see that the models perform less effectively for poorly represented types. The results are usually higher for well represented genres and, within each genre, higher for well represented types. By looking at the uncertainty distribution for both models, we can see that the majority lies in a range at a low confidence (with a small improvement in the multimodal model). The models aren't assigning types with a great certainty, and it was already the case for the genre classification. A telling example: *parody*, which is the less represented type (with only 19 samples), is never predicted well. In the opposite, another poorly represented type, *pantomime*, has an accuracy of 60% for the unimodal model that jumps to 100% in the multimodal one. It's difficult to explain the exact reason, but we can again fear some data leakage.

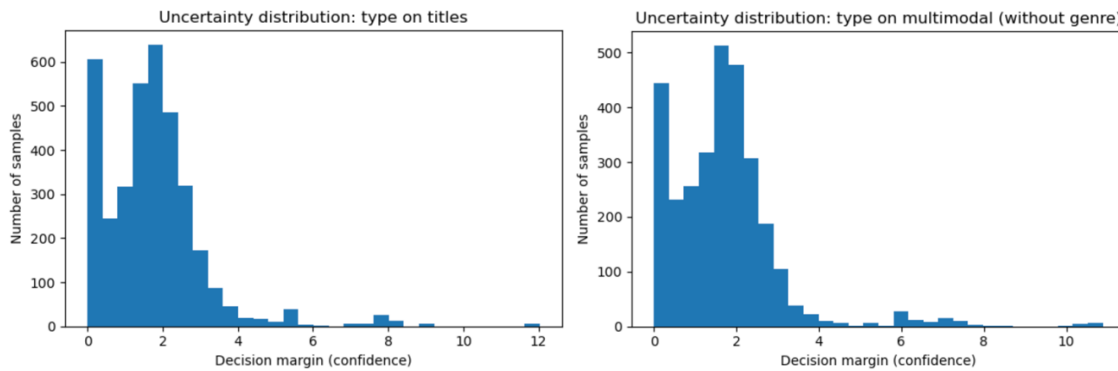


Figure 4: Uncertainty distributions for type based on unimodal and multimodal models

c. Type: multimodal models, predicted genres (*model 5*) vs gold genres (*model 6*)

The models are once more run on a 15 091 samples data frame, with one more time 80% used for training, and 20% for testing. The model trained on predicted genres has a final test accuracy of 90.63%, which is a small improvement with regards to the multimodal model trained on no genre. Confronting this model to the gold-genres one, we only lose less than one percent of accuracy, as the last model performs with an accuracy of 91.32%.

Once again, *parody* was never accurately classified, and was both times considered as *story*, a largely spread subgenre within the same genre class. We can here easily see the weight that overly represented types per genre represents. At this point, it would also be interesting to have a closer look at the *parody* titles, to try to understand the reason why they're never correctly predicted.

7. Discussion

The differences of accuracy in using one or another model are not huge, but they still exist. They only amount to two or three percent and it is hard to tell whether they are due to a real improvement of the model or potential data leakage caused by a bias towards heavily represented classes. Even if we can observe a decrease of uncertainty when the models get more complex, none of them is largely confident about its predictions.

If it can be shown that data leakage wasn't the cause of improvement of more complex models, then using a multimodal one would be more beneficial, as well as taking predicted genre into account when assigning a type (as this model is nearly as good as the one with gold-labeled genres).

Still, assigning types automatically would lead to 10% of errors, which is not negligible. A better idea would be to isolate cases where uncertainty margins are large, to only predict these genres and types and thereby reducing the number of misclassifications. By doing that, one should take more time to evaluate the models and understand why they're predicting well/badly and with which confidence level.

As we have seen in the part of descriptive statistics, some genres and types are largely represented throughout the dataset. It is clear that good results in these areas will boost the overall accuracy, sometimes hiding misclassification issues on underrepresented genres and types.

There are many ways to build machine learning models, and we surely don't pretend to cover all the field. Still, it is reassuring to know that such simple models don't work that bad when based only on small text lengths and side information such as author, place, and date of publication.

8. Conclusion

During this investigatory work about how would perform a machine learning model to assign genres and types to unseen data, we have seen that if the test accuracy scores are not that bad (and especially for the genre attribution), they hide the fact that a correct classification mostly works for mainly represented data. We here fall into one of infamous biases raised by artificial intelligence, that is, overrepresentation of abundant groups. Moreover, the choice of only working with English data, chosen by convenience and by need of enough samples, sadly falls into this representation bias.

The models present encouraging results, but sometimes lack interpretability without further analyses (that weren't all covered here). Conducting them would allow to enhance their functioning. It is also needless to say that the accuracy of our models would only work with a previous proper human annotation and a correct data cleaning.

As it was raised in the project *Classifying 19th Century British Library books using Crowdsourcing and Machine Learning*, we might still initially consider automatic annotation in cases where the machine assigns a high confidence to correct classifications, and so decreasing the amount of work for human annotators. Attributing a book with only one genre label is a difficult and tricky task, even for academics, so it seems fair that the most complex cases remain in specialists' hands.

9. References

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